DEEP-LEARNING VISION INSPECTION SOFTWARE

IntLearn Pro

Deep learning inspection software that finds defects like an expert

12 engines and modules, the most in Korea

Direct it in plain words (LLM link) · carry the inspection in one file (.izpkg)



Segmentation concept diagram · layer by layer, only the defect's shape is left

02 — MANIFESTO

The eye for defects is learned from experts and kept on the line.

IntLearn Pro is deep learning inspection software that finds defects like an expert, with 12 engines and modules, the most in Korea, and LLM integration.

Experts mark OK and NG on photos; the AI learns those calls and judges everything from film to PCBs on the line by the same criteria.

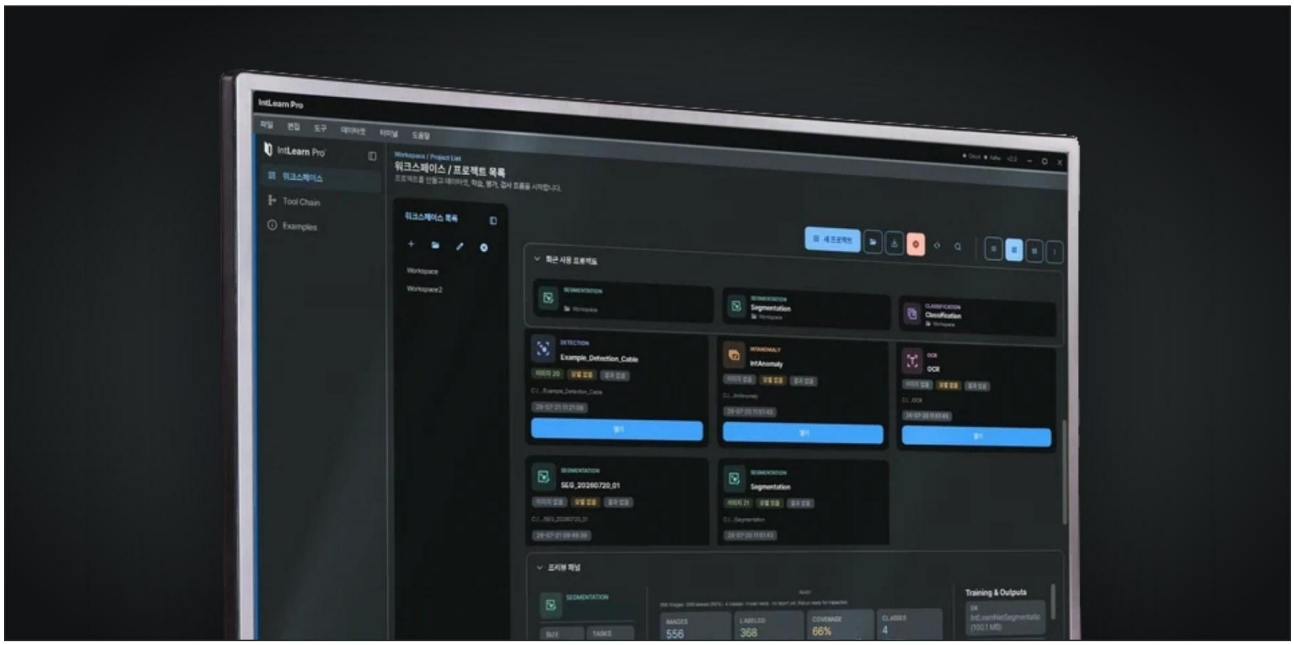
INTELLIZ

03 — OVERVIEW

Find defects with AI, carry the inspection in one file

IntLearn Pro turns experts' OK/NG calls into training data and inspects film to PCBs with six AI engines and six modules INTELLIZ has tuned for the factory floor, all in one program. You direct it in plain words through the IntWise LLM, and the models and criteria go to the line in one file: an inspection package (.izpkg).

FIG. 01 · INTLEARN PRO SCREEN (KOREAN UI) SOFTWARE



12	62	LLM	Patent
Engines and modules	**Tool Chain tools**	**Linked with IntWise**	**AI inspection validation**
6 AI engines + 6 modules	Linked like blocks	Direct it in plain words	Registered

12 engines and modules, the most in Korea in one program

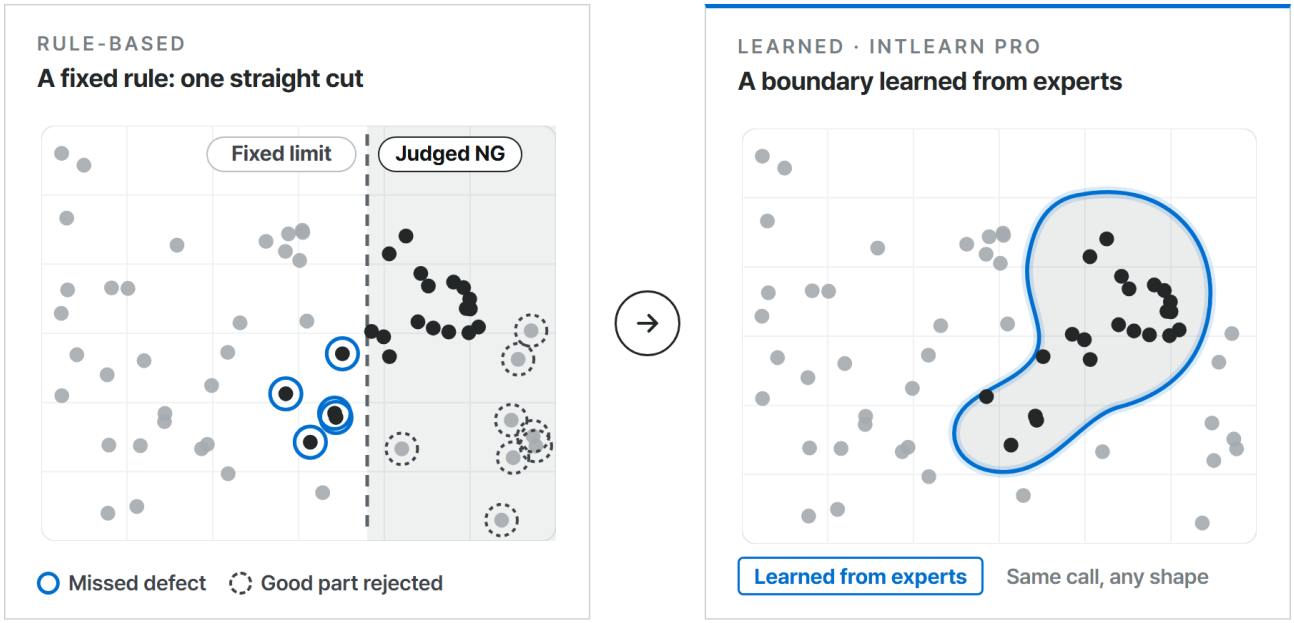
6 AI engines	Anomaly detection · segmentation · OK/NG classification · detection · change comparison · character recognition (OCR)
6 modules	Defect synthesis · augmentation · image enhancement · alignment · label assist · measurement
Linked	LLM integration with IntWise · optical control with IntCloud Optic

04 — WHY AI VISION

Still leaving inspection to a few fixed numbers?

Rules built from a few fixed limits on brightness or size (thresholds) only catch the defect shapes someone expected. Real defects change shape and position every time: tighten the rules and good parts are thrown away, loosen them and defects slip through.

FIG. 02 · CONCEPT ● Good (OK) ● Defect (NG)



01

Misses · defects get out

Defects outside the rules ship as they are and come back as customer claims.

02

Overkill · good parts lost

Anything that looks slightly different is judged NG, adding staff for re-checks and scrap cost.

03

Expert-bound · calls vary

In the end a person looks again. Calls change with the inspector and the shift, and the know-how is not kept.

With IntLearn Pro, nobody writes rules by hand.

It learns the OK/NG photos experts chose, and the AI draws the line between good and defective.

05 — WHAT IS INTLEARN PRO

The expert's eye, as AI on the line

IntLearn Pro is deep learning vision inspection software that turns the OK/NG calls of people on the floor into training data and finds defects by the same criteria. Instead of writing rules in code, you mark photos and train, and inspect everything from film to PCBs the same way.

FIG. 03 · EXPERT → AI → LINE CONCEPT



An expert marks the photo
OK/NG calls marked on the photo become training data as they are

AI learns the difference
It finds the line between good and defective parts in the photos by itself

Judged on the line
Sent in one package; IntScan Pro and IntSight judge by the same criteria

What changes with IntLearn Pro

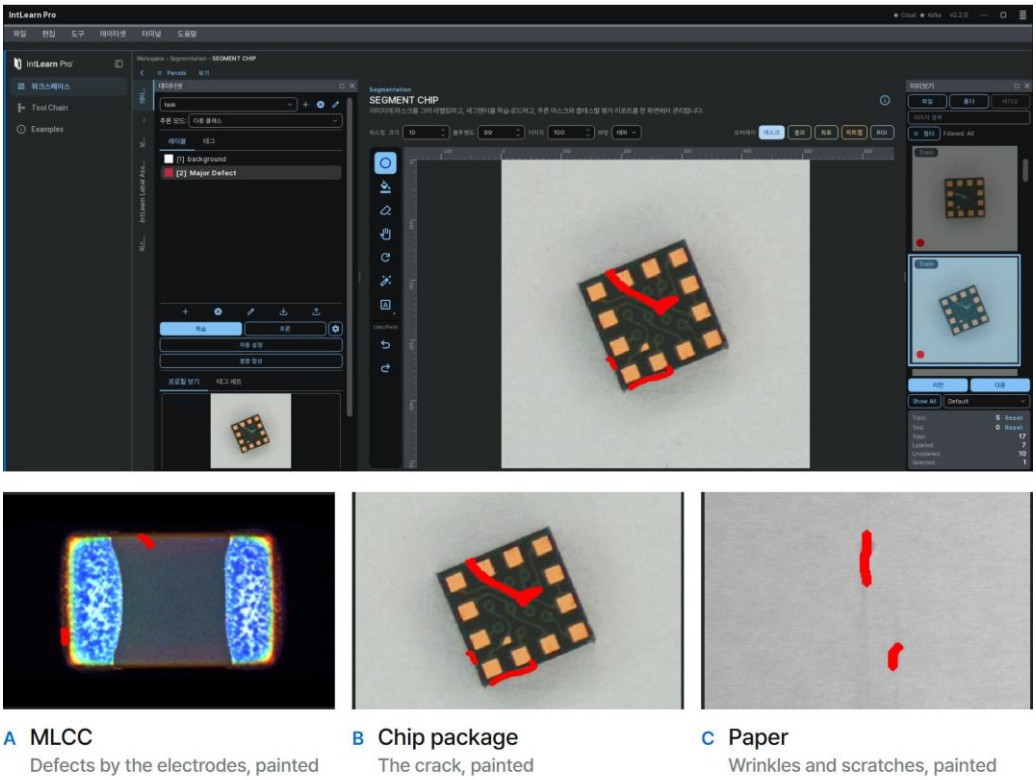
Before		IntLearn Pro
People design limits and rules by hand	→	Experts mark OK/NG on photos, and the AI learns the criteria
Waits for enough defect samples	→	Start from 4 good images (anomaly detection: learns good parts only and flags anything unusual), and add defect images by copying real defects to new places (defect synthesis)
Each line or plant starts over	→	One inspection package (.izpkg) carries the same flow and criteria

06 — END-TO-END WORKFLOW

From the first photo to the line, in one program

No switching between programs: look at the results, add photos and train again, all on one screen. The finished inspection is exported as a model file (.dx) and an inspection package (.izpkg).

FIG. 04 · SEGMENTATION SCREEN · KOREAN UI — Red = defect painted by a person



A MLCC
Defects by the electrodes, painted

B Chip package
The crack, painted

C Paper
Wrinkles and scratches, painted

01

Learns expert criteria

From labeling (marking defects on photos) to augmentation, training and validation, on one screen.

02

Auto labeling

On the images you select, the trained model drafts the labels; a person checks and confirms them.

03

Evaluation reports

HTML reports with miss and overkill rates and accuracy scores (mAP, IoU, AUROC and more).

04

Inspection package

Puts the model, inspection chain and limits in one file (.izpkg) for the line, with versions.

05

Fast runtime conversion

Converts detection models to a fast GPU format (IntLearnRT), packed in the model file (.dx).

06

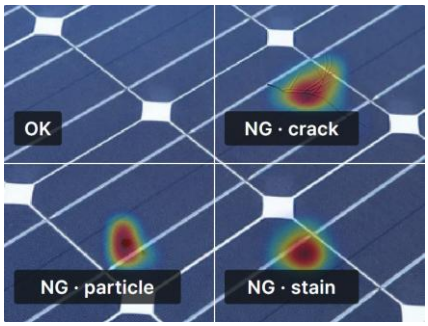
On-premises

Runs on a Windows PC in your plant; in the basic setup, production data stays on site.



Six AI engines, six ways of looking at defects

'Finding defects' asks different questions: what is it, where is it, what shape is it, what does it say, is it unusual, is it different from the reference? Pick the engine that fits the question.



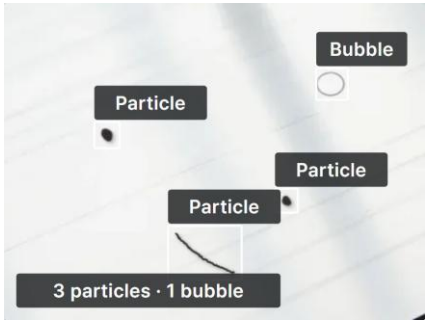
T-01 IntClassification

Classification

A name tag for every photo

Decides OK or the defect type (crack, foreign matter, stain) and shows, as a color map (heatmap), where the AI looked to decide.

2 models · speed first / accuracy–speed balance



T-02 IntDetection

Detection

Where, and how many? Shown with boxes

Boxes show where foreign matter and bubbles on film are, and how many.

4 detection engines · 19 model sizes



T-03 IntSegmentation

Segmentation

Paints the defect's exact shape

Paints defects such as cracks pixel by pixel, as an area of exactly their shape (a mask).

7 architectures · 10 backbones (the part that picks out image features)

08 — AI ENGINES · T-04 — T-06



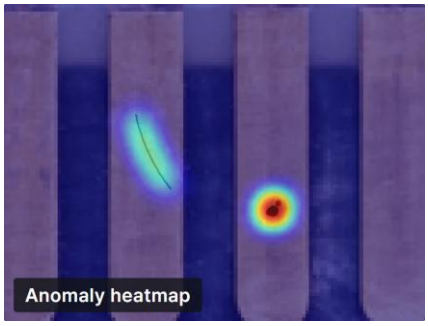
T-04 IntOCR

OCR

Reads engraved and printed text

Finds part markings and engraved text, and reads every line automatically, tilted ones included.

2 recognizers · 7 text detectors · 5 training types



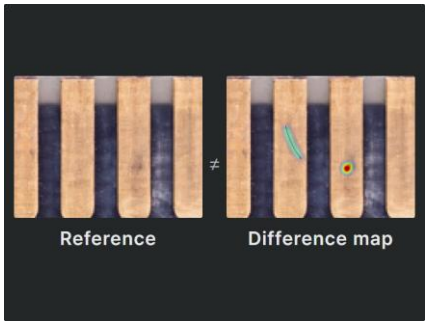
T-05 IntAnomaly

Anomaly detection

Learns good parts only; anything different is flagged

Trains on good images only and finds even unseen foreign matter with an anomaly heatmap and score, then decides OK/NG.

2 methods (memory bank · deep reconstruction) · from 4 good images



T-06 IntComparison

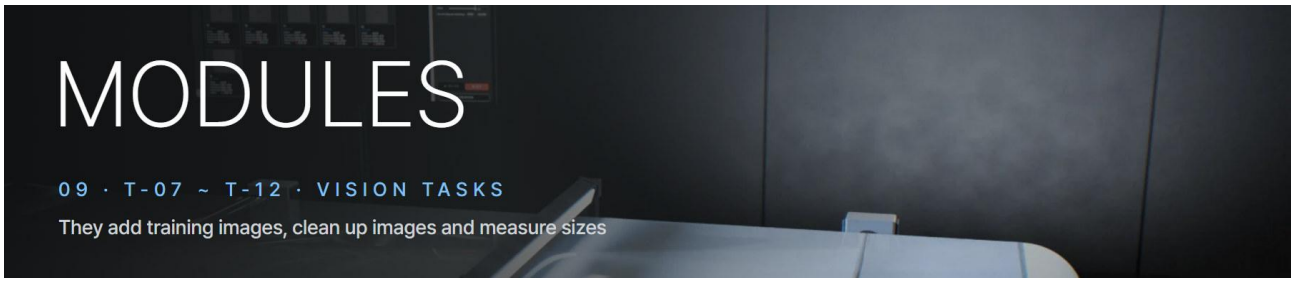
Change comparison

Spot the difference against a reference part

Compares reference and test images by features the AI extracts, and shows what changed as a difference map.

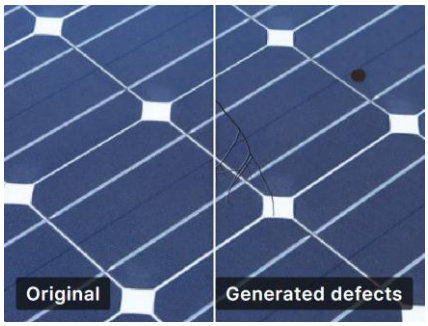
Several reference images · 4 feature backbones · difference map

The defects, heatmaps, masks and boxes in the examples are generated and drawn on real product photos for illustration.



Six modules, for little data and poor images

On the line the real problems are ‘we have no defect samples’, ‘the images are dark and crooked’, ‘we must measure sizes too’. The six modules take these on one by one.



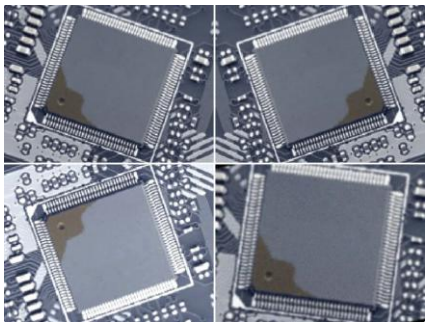
T-07 IntSynthesis

Defect synthesis

Copy and paste to add defects

Copies real defects a person marked to new positions to add NG training images, and makes their labels automatically.

4 blend modes · 4 placement modes · optional AI refinement



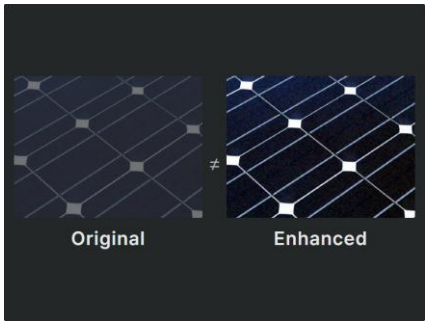
T-08 IntAugmentation

Augmentation

Flip and turn to practise ahead

Varies flips, rotation, brightness, noise and more while training, for a model that holds up to changes on the line.

11 transforms · auto-tune · copy multiplier



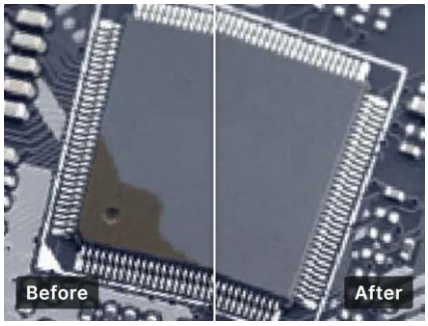
T-09 IntEnhancement

Image enhancement

Makes images clear

Corrects dark, low-contrast images so they are easier to inspect.

Contrast, brightness, lighting · sharpening · denoising · text-image correction



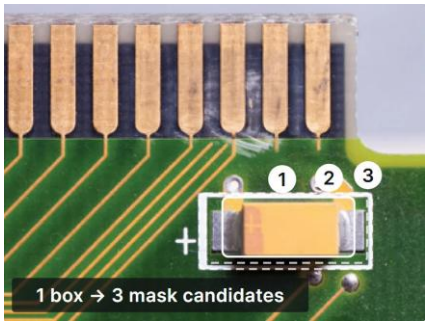
T-10 IntAlignment

Alignment

Straightens parts that come in crooked

Fixes position, rotation and skew to the reference before inspection.

4 modes (translation, rotation, affine, perspective) · position fix · auto text deskew



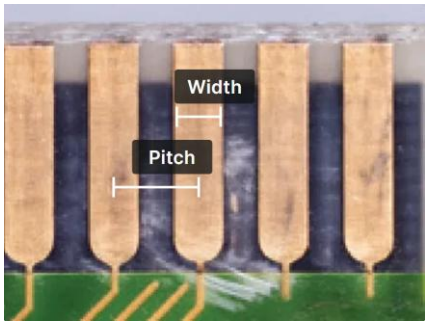
T-11 IntLabelAssist

Label assist

Draw one box; the AI does the rest

Draw one box and get three candidate areas to paint (masks), cutting the time spent painting defects pixel by pixel.

5 label-assist models, from fast to high capacity



T-12 IntMeasurement

Measurement

Measures right where it found the defect

Measures length, width, gap, circularity and more from segmentation results and judges them against limits.

26 measurement tools · judged against limits

The defects, heatmaps, masks and boxes in the examples are generated and drawn on real product photos for illustration.

Choose by the problem on the line

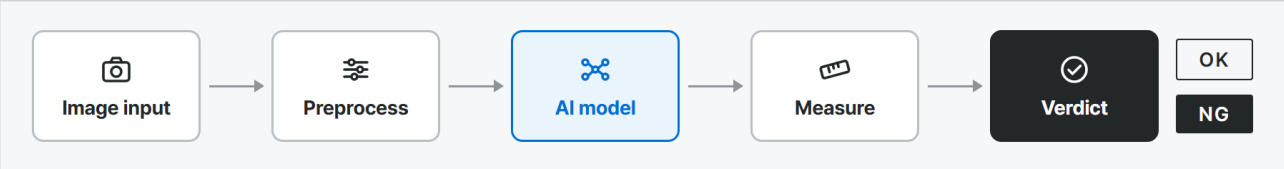
Problem on the line		Engine or module	How it is solved
Hardly any defect samples	→	Anomaly detection · defect synthesis	Start from 4 good images, and copy real defects to add NG images
Changes vs. a reference part	→	Change comparison	Compare with several reference images
Dark or tilted images	→	Enhancement · alignment	Correct them, then train and inspect
Sizes and gaps to measure	→	Segmentation · measurement	Find the area, then measure it in place

11 — TOOL CHAIN · INTEGRATION

Pick tools for the problem, link them like blocks

Inspection does not end with one AI. Taking the image, cleaning it up, finding with AI, measuring and deciding are linked like blocks into one inspection flow (Tool Chain).

FIG. 05 · TOOL CHAIN BLOCKS



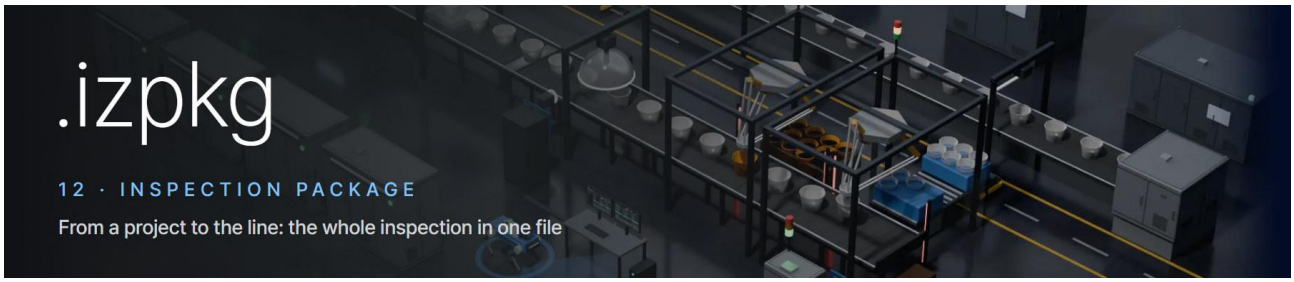
62 Tool Chain tools By category · chain validation · input mapping · judgment profile (pass criteria) · verdict explanation

8	12	12	8	12	5	5
Position · alignment	Preprocessing · enhancement	Defect detection · regions	Deep learning models	Measurement	Combine	Output

FIG. 06 · INTEGRATION LLM · OPTICS







Inspection know-how in one file: the .izpkg

Trained models, the inspection flow and pass criteria go into one file for the line: IntScan Pro (line scan: captures moving material line by line) and IntSight (area scan: one frame per part). Like an app installer, it carries the same flow and criteria to other lines and plants, within its compatible range.

FIG. 07 · INSPECTION PACKAGE inspection.izpkg · ZIP



01 BUILD

Name, version

Exported with a name and a version

02 VERIFY

Golden set

Known-answer images (golden set, optional) · integrity check

03 RUNTIME

Lightweight

No training data or checkpoints (saved training states)

04 RECORD

SHA-256, history

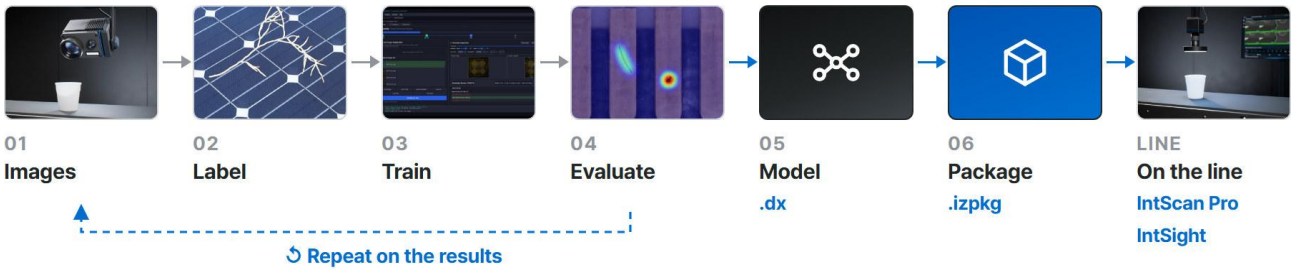
SHA-256 file fingerprint; apply and rollback log per site

13 — FROM PROJECT TO LINE

From a project to the line, with the same criteria

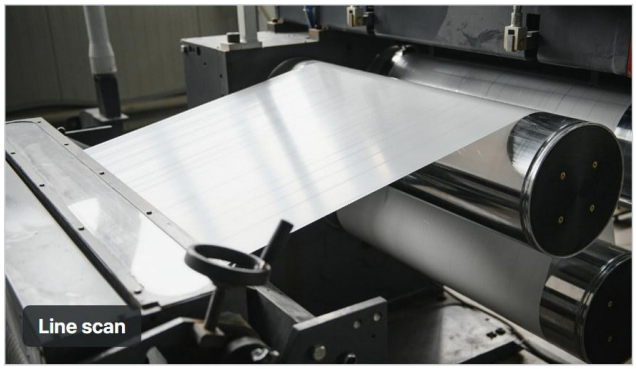
In a project you repeat photos → labeling (marking defects) → training → evaluation while watching the results, then pack the finished model (.dx) with the inspection flow and pass criteria into a package (.izpkg) and send it to the line.

FIG. 08 · FROM A PROJECT TO THE LINE PROJECT → LINE

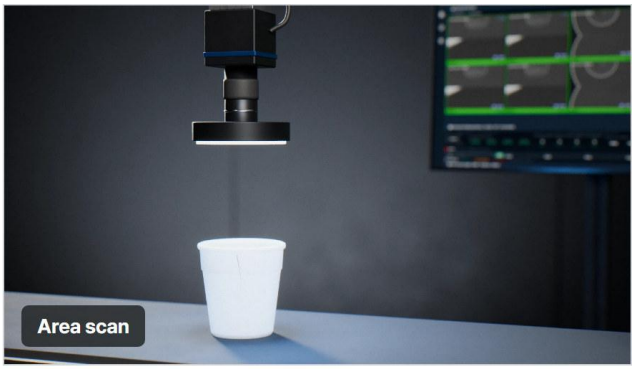


On the line

Line scan IntScan Pro · area scan IntSight



IntScan Pro · film or paper, captured line by line as it moves



IntSight · each part captured in one frame, then judged

Line scan or area scan, the same package.

IntScan Pro and IntSight both take the inspection flow and pass criteria from the same single .izpkg.

14 — EVIDENCE · CASES

Results in numbers

2.47 ms
External test average[1]
Segmentation engine only · 1024×1024

59.5%
Shorter processing time[2]
Secondary-battery cans 200 → 81 ms

1.4×
Inspections per minute[2]
MLCC 2,934 → 4,100

About 1/3
Implementation cost[2]
Versus existing equipment

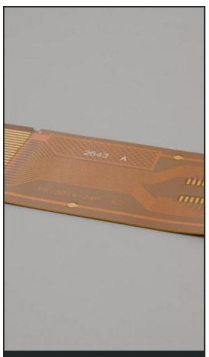
1) The average measured with the segmentation engine on 1024×1024 images in the KIPT (Korea Institute of Testing) deep learning inspection test (KIPT-QT2026-008) · 100% repeatability. It is the engine's processing time, not the total inspection time or the line's throughput.
2) 59.5%, 1.4× and about 1/3 are company comparisons and can vary with the conditions of each site.

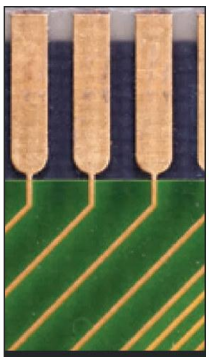
Industries

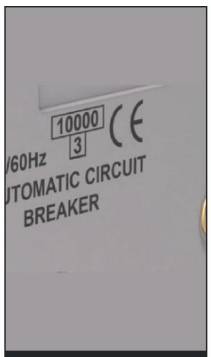
Film · paper · FPCB · electronic PCB · power equipment · electronic components

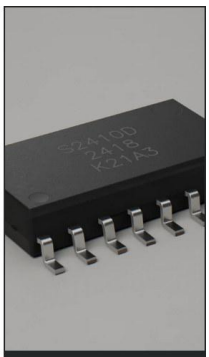












INTELLIZ use cases

Consumer goods, plastic & containers · The scenes are 3D renderings, shown as examples

CASE 01 · VIDEO

Defect inspection of injection-molded containers

Defects: deformation · cracks · burrs · short shots · flash · dimensional errors

CASE 02 · VIDEO

Defect inspection in film manufacturing

Defects: bubbles · scratches · foreign matter

CASE 03 · VIDEO

Cleaning inspection of reusable containers

Defects: residual contamination (foreign matter, grease, water marks) · damage

15 — VIDEO · LINKS

Watch, then find out more

Watch the videos

Point your phone camera at a QR code

IntLearn Pro · INTELLIZ
From expert judgment to an inspection package

Technical Series #1
System overview

Technical Series #2
6 project models

Technical Series #3
End-to-end workflow

YouTube @intelliz
More on the YouTube channel

More at intelliz.net

Try it	Try IntLearn Pro in a virtual copy of a factory (digital twin)	intelliz.net/en/solutions/experience?station=st-02
Line scan	See IntScan Pro, the line-scan inspection runtime	intelliz.net/en/products/intscan-pro
Area scan	See IntSight, the area-scan inspection runtime	intelliz.net/en/products/intsight
Optics Lab	Can line scan or area scan inspect your product?	intelliz.net/en/solutions/optics-lab

Products linked on the platform



IntCloud Optic

An AI optical inspection module that captures defects so they can be seen



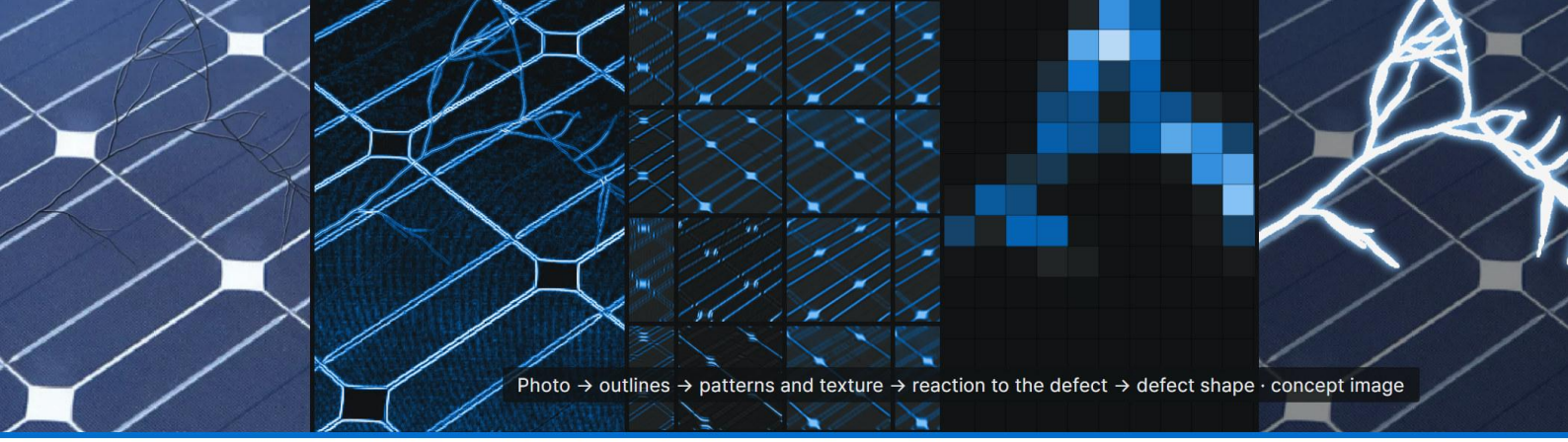
IntLearn Kinesis New in 2026

Validate in a virtual factory before you install



IntWise New in 2026

Operate by speaking, and the know-how becomes a company asset



16 — CONTACT · NEXT STEPS

Considering a similar inspection?

Including the following in your inquiry helps us prepare our answer.

01	What you inspect: material, shape and size	02	The defects to find and your pass/fail criteria
03	Line speed and how products are conveyed	04	Space available for cameras and lighting
05	Where results need to go (PLC, reject unit, etc.)		

Next steps

Links on intelliz.net

Inspection guides

AI Inspection Metrics: Missed Defects and False Rejects

intelliz.net/en/guides/ai-inspection-evaluation-metrics

CEO Na's Defect Clinic

Only 3 Defect Samples? AI Inspection That Learns From Good Parts

intelliz.net/en/solutions/defect-clinic/ep-06

CEO Na's Defect Clinic

Catching NGs but Not the Cause? How to Pick an AI Defect Classifier

intelliz.net/en/solutions/defect-clinic/ep-02

Getting ready

AI Inspection Adoption Checklist

intelliz.net/en/tools/inspection-checklist

FAQ

What does IntLearn Pro do?

intelliz.net/en/company/faq#intlearn

CONTACT

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Product page

Try it

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Main Office #305, 63, Buldang 36-gil, Seobuk-gu, Cheonan-si, Chungcheongnam-do, Republic of Korea
Manufacturing Lab B-403, Hyundai Silicon Alley Dongtan, 150, Dongtanyeongcheon-ro, Hwaseong-si, Gyeonggi-do, Republic of Korea
Japan distributor BIT DEVICE Co., Ltd., 6-3-24-L430 Nishinakajima, Yodogawa-ku, Osaka, Japan

INTELLIZ

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